

- Introduction:

Probability: Probability is the branch of mathematics or statistics that deals with the concept and measurement of uncertainty and randomness.

The theory of probability provides methods of quantifying the chances or the likelihood associated with the various outcomes

Uses of Probability

Probability provides tools for handling statistical analysis in connection with insurance, elections, health, finance, gambling, economic/business, opinion polls, quality control, etc.

Set Operation, Venn Diagrams and Theorems on Sets

The concept of a set:

At the foundation of probability and Statistics and of mathematics in general, is the concept of a set. A set is a collection of objects, called member or elements of the set. In general, unless otherwise specified, we denote a set by a capital letter such as A, B, C , and an element by a lower case letter such as a, b , etc.

The synonyms for set are class, aggregate and collection.

If an element a belongs to a set C , we write $a \in C$. If a does not belong to C , we write $a \notin C$. If both a and b belong to C , we write $a, b \in C$.

A set can be defined by actually listing its elements (roster method) or by describing some property held by all members and by no nonmembers.

Example: The set of all vowels in English alphabet can be defined by

① roster method as $V = \{a, e, i, o, u\}$

② property method $V = \{v \mid v \text{ is a vowel}\}$.

Subsets

If each element of a set A also belongs to a set B , we say A is a subset of B , written as $A \subset B$ or $B \supset A$ read as " A is contained in B " or " B contains A ".

If $A \subset B$ and $B \subset A$, we call A and B equal and we write $A = B$. In such cases A and B have exactly the same elements.

If A is not equal to B , i.e. if A and B do not have exactly the same elements, we write $A \neq B$.

If $A \subset B$ but $A \neq B$, we call A a proper subset of B .

Example: Is $\{a, i, u\}$ a proper subset of $\{a, c, i, o, u\}$?

Solⁿ: Yes because $\{a, i, u\} \subset \{a, c, i, o, u\}$
and $\{a, i, u\} \neq \{a, c, i, o, u\}$.

Example: Is $\{i, a, o, u, e\}$ a proper subset of $\{a, e, i, o, u\}$?

Solⁿ: No, because the two sets are equal

Example: In tossing a die, the possible outcomes where the die comes up "even" is a proper subset?

Solⁿ: In tossing a die, the possible outcome is $\{1, 2, 3, 4, 5, 6\}$
possibility of even number is $\{2, 4, 6\}$

$\{2, 4, 6\} \subset \{1, 2, 3, 4, 5, 6\}$ and $\{2, 4, 6\} \neq \{1, 2, 3, 4, 5, 6\}$

So the event of getting even numbers is proper subset.

Theorem:

if $A \subset B$ and $B \subset C$, then $A \subset C$.

(4)

Universal Set and Empty Set

For the ^{many} purposes we restrict our discussion to subsets of some particular set called the universe of discourse or briefly universe or universal set or space denoted by U .

The elements of a space are called points of the space.

Empty Set ϕ , is a set having no elements at all. It is also known as null set

Empty or null set is a subset of any set

Closed Intervals: Consider the set $\{x \mid 2 \leq x \leq 5\}$

Such a set is closed interval.

The elements are $\{2, 3, 4, 5\}$

Open Intervals: Consider the set $\{x \mid 2 < x < 5\}$

Such a set is open interval.

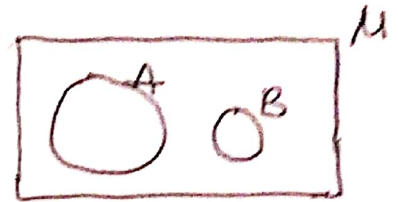
The elements are $\{3, 4\}$

Half open or half closed interval: Subset such as

$\{x \mid 2 \leq x < 5\}$ or $\{x \mid 2 < x \leq 5\}$. (5)

VENN DIAGRAMS

A universe U can be represented geometrically by the set of points inside a rectangle. In such case



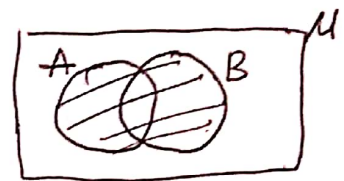
subsets of U i.e. A and B

shown in the figure beside are represented by circles.

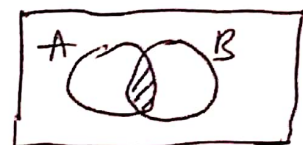
Such diagrams called Venn diagrams are used to serve as geometric intuition regarding possible relationships between sets.

Set Operations

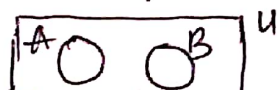
Union: The set of all elements (or points) which belong to either A or B or both A and B is called union of A and B , denoted by $A \cup B$



Intersection: The set of all elements which belong to both A and B is called intersection of A and B , denoted by $A \cap B$.

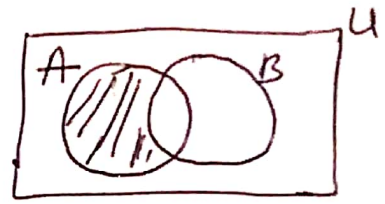


Disjoint: Two sets A and B such that $A \cap B = \emptyset$ are known as disjoint set



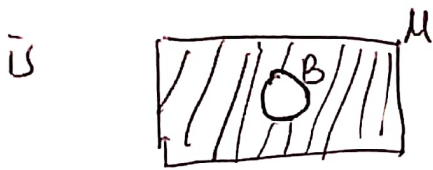
(6)

Difference: The set consisting of all elements of A which do not belong to B is called the difference of A and B denoted by $A - B$.



Complement

Given U , the complement of B denoted by B'



$$B' = U - B$$

Note also that the complement of $A \cup B$ is denoted by $(A \cup B)'$

Theorems involving Sets

Theorem 1: $A \cup B = B \cup A$ Commutative law of unions

Theorem 2: $A \cup (B \cap C) = (A \cup B) \cap C = A \cup B \cap C$ Associative law of unions

Theorem 3: $A \cap B = B \cap A$ Commutative law of intersections

Theorem 4: $A \cap (B \cup C) = (A \cap B) \cup C = A \cap B \cup C$ Associative law of intersections

Theorem 5: $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ Second distributive law

Theorem 6:

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C) \quad \text{First distributive law}$$

Theorem 7: $A - B = A \cap B'$

Theorem 8: if $A \subset B$, then $A' \supset B'$ or $B' \subset A'$

Theorem 9: $A \cup \phi = A$, $A \cap \phi = \phi$

Theorem 10: $A \cup \mu = \mu$, $A \cap \mu = A$

Theorem 11: $(A \cup B)' = A' \cap B'$ De Morgan's first law

Theorem 12: $(A \cap B)' = A' \cup B'$ De Morgan's second law

Theorem 13: $A = (A \cap B) \cup (A \cap B')$ for any set A and B.

Cardinality of a set

The number of elements present in a set.

Example: if $A = \{4, 8, 16, 32, 64, 128\}$, then the cardinality of set A is $n(A) = |A| = 6$

Example: if $C = \{\text{all even numbers from 1 to 15}\}$ then the cardinality of set C is $n(C) = |C| = 7$. (8)

Example: find $|\{2, 6, 7\}|$.

S/n $|\{2, 6, 7\}|$ is 3.

If $A = \phi$, then the cardinality of set A is

$$n(A) = |A| = 0$$

Finite and Infinite Sets

A finite set contains a limited, countable number of elements, while an infinite set contains uncountable number of elements, continuing indefinitely.

Examples of finite sets

① The set of days in a week:

{Monday, Tuesday, Wednesday, Thursday, Friday, Saturday, Sunday}

② The set of vowels in the English alphabet

{a, e, i, o, u}

Examples of infinite sets

① The set of all natural numbers: $\{1, 2, 3, 4, \dots\}$

② Set of all integers: $\{\dots, -2, -1, 0, 1, 2, \dots\}$

Exercise

(1) Describe the following in words

- (a) $A \cup B$ (b) $A \cap B$ (c) A^c (d) $A - B$

Solution

(a) $A \cup B$ is the event "either A or B or both"

(b) $A \cap B$ both A and B

(c) A^c is the event "not A"

(d) $A - B$ is the event "A but not B"

(2) If the universe $U = \{\frac{1}{2}, 0, \pi, 5, -\sqrt{2}, -4\}$ and the subsets of U are $A = \{-\sqrt{2}, \pi, 0\}$, $B = \{5, \frac{1}{2}, -\sqrt{2}, -4\}$ and $C = \{\frac{1}{2}, -4\}$ find:

(a) $A \cap B$ (b) $A \cup B$ (c) $(A \cup B) \cap C$ (d) $B^c \cup C^c$

(e) $A - B$ (f) $(B \cap C)^c$ (g) $(A \cap C) \cup (B \cap C)$ (h) $|U|$

Solution

$$(a) A \cap B = \{-\sqrt{2}, \pi, 0\} \cap \{5, \frac{1}{2}, -\sqrt{2}, -4\} = \{-\sqrt{2}\}$$

$$(b) A \cup B = \{-\sqrt{2}, \pi, 0\} \cup \{5, \frac{1}{2}, -\sqrt{2}, -4\} = \{-\sqrt{2}, \pi, 0, 5, \frac{1}{2}, -4\}$$

$$(c) (A \cup B) \cap C = \{-\sqrt{2}, \pi, 0, 5, \frac{1}{2}, -4\} \cap \{\frac{1}{2}, -4\} = \{\frac{1}{2}, -4\}$$

$$(d) B^c \cup C^c = \{0, \pi\} \cup \{0, \pi, 5, -\sqrt{2}\} = \{0, \pi, 5, -\sqrt{2}\} \quad (10)$$

Exercise

① Describe the following in words

(a) $A \cup B$ (b) $A \cap B$ (c) A' (d) $A - B$

Solution

(a) $A \cup B$ is the event "either A or B or both"

(b) $A \cap B$ both A and B

(c) A' is the event "not A"

(d) $A - B$ is the event "A but not B"

② If the universe $U = \{\frac{1}{2}, 0, \pi, 5, -\sqrt{2}, -4\}$ and the subsets of U are $A = \{-\sqrt{2}, \pi, 0\}$, $B = \{5, \frac{1}{2}, -\sqrt{2}, -4\}$ and $C = \{\frac{1}{2}, -4\}$ find:

(a) $A \cap B$ (b) $A \cup B$ (c) $(A \cup B) \cap C$ (d) $B' \cup C'$

(e) $A - B$ (f) $(B \cap C)'$ (g) $(A \cap C) \cup (B \cap C)$ (h) $|U|$

Solution

(a) $A \cap B = \{-\sqrt{2}, \pi, 0\} \cap \{5, \frac{1}{2}, -\sqrt{2}, -4\} = \{-\sqrt{2}\}$

(b) $A \cup B = \{-\sqrt{2}, \pi, 0\} \cup \{5, \frac{1}{2}, -\sqrt{2}, -4\} = \{-\sqrt{2}, \pi, 0, 5, \frac{1}{2}, -4\}$

(c) $(A \cup B) \cap C = \{-\sqrt{2}, \pi, 0, 5, \frac{1}{2}, -4\} \cap \{\frac{1}{2}, -4\} = \{\frac{1}{2}, -4\}$

(d) $B' \cup C' = \{0, \pi\} \cup \{0, \pi, 5, -\sqrt{2}\} = \{0, \pi, 5, -\sqrt{2}\}$ (10)

(c) $A - B$ = set of elements in A which are not in $B = \{0, \pi\}$

or equivalently, using theorem 7,

$$A - B = A \cap B' = \left\{\frac{1}{2}, 0, \pi, 5, \sqrt{2}, -4\right\} \cap \{0, \pi\} = \{0, \pi\}$$

$$\oplus B \cap C = \{5, \frac{1}{2}, -\sqrt{2}, -4\} \cap \{\frac{1}{2}, -4\} = \{\frac{1}{2}, -4\}$$

$$\text{Then } (B \cap C)' = \{0, \pi, 5, -\sqrt{2}\}$$

* We can also use the De Morgan's second law.

$$\ominus A \cap C = \{-\sqrt{2}, \pi, 0\} \cap \{-\frac{1}{2}, -4\} = \phi$$

$$B \cap C = \{-\frac{1}{2}, -4\}$$

$$\text{then } (A \cap C) \cup (B \cap C) = \phi \cup \{-\frac{1}{2}, -4\} = \{-\frac{1}{2}, -4\}$$

(h) $|U| = n(U) = 6$, thus the cardinality of U .

Try

① Let U be the set of all nonnegative and consider the subsets $A = \{x \mid x \text{ is an even integer, } 1 \leq x < 6\}$ and $B = \{x \mid x \text{ is a prime number, } 0 < x \leq 4\}$ Find

$$\textcircled{a} A \cup B \quad \textcircled{b} A \cap B \quad \textcircled{c} A' \cap B' \quad \textcircled{d} |A - B| \quad \textcircled{e} B - A$$

$$\textcircled{f} (A - B) \cup (B - A) \quad \textcircled{g} A' \cup B' \quad \textcircled{h} A \cap B' \quad \textcircled{11}$$

(c) $A - B$ = set of elements in A which are not in $B = \{0, \pi\}$

or equivalently, using theorem 7,

$$A - B = A \cap B' = \left\{\frac{1}{2}, 0, \pi, 5, -\sqrt{2}, -4\right\} \cap \{0, \pi\} = \{0, \pi\}$$

$$\oplus B \cap C = \{5, \frac{1}{2}, -\sqrt{2}, -4\} \cap \{\frac{1}{2}, -4\} = \{\frac{1}{2}, -4\}$$

$$\text{Then } (B \cap C)' = \{0, \pi, 5, -\sqrt{2}\}$$

* We can also use De Morgan's second law.

$$\textcircled{9} A \cap C = \{-\sqrt{2}, \pi, 0\} \cap \{-\frac{1}{2}, -4\} = \phi$$

$$B \cap C = \{-\frac{1}{2}, -4\}$$

$$\text{then } (A \cap C) \cup (B \cap C) = \phi \cup \{-\frac{1}{2}, -4\} = \{-\frac{1}{2}, -4\}$$

(h) $|U| = n(U) = 6$, thus the cardinality of U .

Try

(1) Let U be the set of all nonnegative and consider the subsets $A = \{x \mid x \text{ is an even integer, } 1 \leq x < 6\}$ and $B = \{x \mid x \text{ is a prime number, } 0 < x \leq 4\}$ Find

(a) $A \cup B$ (b) $A \cap B$ (c) $A' \cap B'$ (d) $|A - B|$ (e) $B - A$

(f) $(A - B) \cup (B - A)$ (g) $A' \cup B'$ (h) $A \cap B'$ (11)